MTH6134 Statistical Modelling II

Working as a Group

The group project runs across the whole term. This document explains how to organise your group's work from the start so that you build up a clear record of what you did as you go. Doing this well will make the final write-up much easier and will give you the evidence you need to show how your analysis developed.

Set this up as soon as your group is allocated. Do not leave it until later in the term.

Set up a shared workspace

As soon as your group is allocated, set up a single shared workspace that your whole group can access. All of your group's work should live in this one place. Keep everything here from the beginning; it is much easier to maintain a shared workspace from the start than to assemble one at the end from scattered files.

Any tool that lets everyone in the group read and add to the same set of files will work. Options to consider:

- A version control repository, such as GitHub. This is the option I would encourage: it keeps a full history of changes and who made them, which is exactly the record of who did what that you will need later.
- A collaborative document or notebook, such as Google Colab or a shared RStudio project, which lets several people work on the same code at once.
- A shared cloud folder, such as OneDrive or Google Drive.

Whichever you choose, keeping all your code and comments together in one place as you go will make it much easier to write the report at the end.

You will submit your workspace as part of the final group submission, so keep it tidy and make sure it is self-contained: include the data or the code to read it, use relative file paths rather than paths specific to one person's computer, and check that everything runs from the folder as submitted. This avoids technical problems at the end and gives clear evidence of how your group worked through the term.

Keep all of your work

Your workspace should accumulate everything your group produces, not just the polished final version. Keep:

- **All code**, including rough exploratory work. Do not delete code just because it did not end up in the final report. The models you tried and discarded are part of the story of how you reached your final analysis.
- **Plots and outputs** you produced along the way, even ones you did not use.
- **Notes on decisions**, in a file named `decision-notes` (for example `decision-notes.docx`

or `decision-notes.md`). When your group makes an analytical decision, write down what you decided, when, and why. For example: which model you chose and what alternatives you considered, how you handled an unusual observation, or why you included or excluded a variable.

- **A record of who did what**, in a file named `who-did-what` (for example `who-did-what.docx` or `who-did-what.md`). Keep a simple running record of which group members worked on which parts of the analysis. This does not need to be formal, but it should be accurate and kept up to date.

Organise your workspace into these subfolders, so it is easy to navigate:

- `code`: all your R code, including rough exploratory work.
- `outputs`: plots and outputs you produced along the way.
- `notes`: your `decision-notes` and `who-did-what` files.
- `reports`: the final group report and each member's individual report.

Why this matters

Keeping this record as you go serves several purposes. It makes the final report easier to write because the reasoning is already documented. It helps you divide the work fairly and see at a glance what has and has not been done. And it provides evidence of how your group worked and how each member contributed, which you will draw on later in the term.

Check-in points

Set up your shared workspace by the end of **week 5**. A submission point for recording its link will be made available on QM+ before then, and you will be notified when it is open; submit your workspace link there by the end of week 5. Make sure the workspace is shared with my QMUL email *e.matechou@qmul.ac.uk* (or is otherwise accessible to me), and update the link if it ever changes.

I will look at each group's workspace at three points in the term, in **weeks 5, 8 and 11**, to see that it is set up and to check how your group is getting on.

Your workspace is not a separately marked item, but I use it when marking: to confirm the report is the group's own work and to check the individual contribution statements against your record of who did what. Keeping it current is therefore in your interest.

Telling me about problems

If your group runs into difficulties, in particular a member who is not engaging or contributing, tell me as soon as possible, not at the end. Problems are much easier to deal with while there is still time to act.

Producing the report

The final report must be submitted as a PDF, but you are free to choose how you produce it. Options include Word, R Markdown (which lets you combine your code, output, and writing in one document), or LaTeX (for example through Overleaf, which supports collaborative editing). R Markdown is well suited to this kind of report because it keeps the analysis and the write-up together and makes the results reproducible, but any tool that produces a clear PDF is acceptable.